

Probability Cheatsheet

concise reference: pdf/pmf, cdf, mean, var, mgf, and use-cases

Probability Space & Basics

$(\Omega, \mathcal{F}, P)$; $P(\Omega) = 1$.
Conditional: $P(A|B) = P(A \cap B)/P(B)$.
Independence: $A \perp B \iff P(A \cap B) = P(A)P(B)$.

Discrete RVs — PMF/CDF/Moments

PMF: $p_X(x) = P(X = x)$, $\sum p_X(x) = 1$.
CDF: $F_X(x) = P(X \leq x) = \sum_{t \leq x} p(t)$.
 $E[X] = \sum xp(x)$, $\text{Var} = E[X^2] - E[X]^2$.
Tail-sum (nonneg int): $E[X] = \sum_{k \geq 0} P(X > k)$.

Continuous RVs — PDF/CDF/Moments

PDF: $f(x) \geq 0$, $\int f = 1$.
CDF $F(x) = \int_{-\infty}^x f(t)dt$, and $f = F'$.
 $P(a < X < b) = \int_a^b f(x)dx$, $E[X] = \int xf(x)dx$.

Bernoulli / Binomial / Poisson (compact)

Bernoulli(p): $P(1) = p$, $E = p$, $\text{Var} = p(1-p)$, $\text{MGF} = 1 - p + pe^t$.
Binomial(n, p): $\binom{n}{k} p^k (1-p)^{n-k}$, $E = np$, $\text{Var} = np(1-p)$, $\text{MGF} (1-p + pe^t)^n$.
Poisson(λ): $e^{-\lambda} \lambda^k / k!$, $E = \lambda$, $\text{Var} = \lambda$, $\text{MGF} \exp(\lambda(e^t - 1))$.

Geometric / NegBin / Hypergeometric

Geometric(p) (trials to 1st success): $p(1-p)^{k-1}$, $E = 1/p$, $\text{Var} = (1-p)/p^2$, memoryless.
NegBin(r, p): $\binom{k-1}{r-1} p^r (1-p)^{k-r}$, $E = r/p$.
Hypergeom(N, K, n): $\frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$, use for sampling without replacement.

Uniform / Exponential / Gamma

Uniform(a, b): $f = 1/(b-a)$, $F = (x-a)/(b-a)$, $E = (a+b)/2$, $\text{Var} = (b-a)^2/12$.
Exponential(λ): $f = \lambda e^{-\lambda x}$ ($x \geq 0$), $F = 1 - e^{-\lambda x}$, $E = 1/\lambda$, $\text{Var} = 1/\lambda^2$, $\text{MGF} \lambda/(\lambda - t)$.
Gamma(α, β): $f = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, $E = \alpha/\beta$, $\text{Var} = \alpha/\beta^2$, $\text{MGF} (\beta/(\beta - t))^\alpha$.

Beta / Normal / Chi-square

Beta(α, β): $f = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)}$, $E = \alpha/(\alpha + \beta)$.
Normal(μ, σ^2): $f = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$, $E = \mu$, $\text{Var} = \sigma^2$, $\text{MGF} e^{\mu t + \frac{1}{2}\sigma^2 t^2}$.
 χ_k^2 : $\text{Gamma}(k/2, 1/2)$, $E = k$, $\text{Var} = 2k$.

t and F distributions

Student-t t_ν : $T = Z/\sqrt{V/\nu}$ with $Z \sim N(0, 1)$, $V \sim \chi_\nu^2$. Heavy tails; use when σ unknown.
F _{d_1, d_2} : ratio of scaled chi-squares; use in ANOVA / variance tests. Relation: $T^2 \sim F_{1, \nu}$.

Transforms, Jacobian, Convolution

Linear: $Y = aX + b \Rightarrow f_Y(y) = \frac{1}{|a|} f_X((y-b)/a)$.
Change of variables (multiv): $f_{U,V}(u, v) = f_{X,Y}(x, y) |\det(J_{T^{-1}})|$.
Ratio: $Z = X/Y \Rightarrow f_Z(z) = \int f_{X,Y}(zw, w) |w| dw$.
Convolution: $f_{X+Y}(z) = \int f_X(x) f_Y(z-x) dx$.

MGF & Useful Identities

$M_X(t) = E[e^{tX}]$, $M^{(n)}(0) = E[X^n]$.
MGF uniqueness: equality on neighborhood of 0 implies same law.
Law of total expectation: $E[X] = E[E[X|Y]]$.
Variance decomposition: $\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$.